

DL for ENG

LECTURE 8.1

Stationary Heat

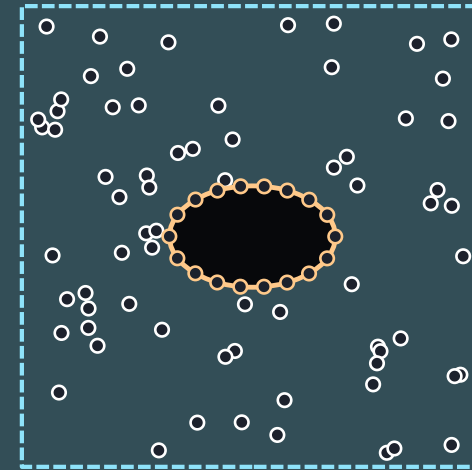
Remus Teodorescu

*With support from Research Assistant Noman Khan -
nomank@energy.aau.dk*

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

CONDUCTION · GEOMETRY · THE PLATE



Steady conduction is Poisson's equation with units: Fourier's law, a conductivity, and walls held at a temperature, a flux or a convection. On a plate with a cooling hole the hole is built into the network by its level set, and the answer is checked by an energy balance.

Learning Objectives

SECTIONS, IN ORDER

- 1 **Heat transfer** — what heat is, its three mechanisms, which L8 solves
- 2 **Conduction** — the steady equation, variable k , the thermal walls, scaling
- 3 **Geometry** — sampling a domain with a hole, its multiplier, its normal
- 4 **Plate with a hole** — verify, solve, weight, and balance the heat

READ, AND WHAT EACH ANSWERS

Incropera ch. 1	how does heat move?
Liu ch. 2.4	what does a thermal problem need?
Liu ch. 3	how is a domain with a hole handled?
Liu ch. 4.6	how is the answer checked?

Recap of L7.2 Fundamental PDEs

1 - 2 • STEADY

1 What decides the type of a second-order equation, and which conditions does each type need?

How it treats time. An elliptic equation has none: a condition on every wall. A parabolic one has one time derivative: the walls and one initial field. A hyperbolic one has two: the walls and two initial conditions. With fewer, the problem has many answers. *Liu ch. 2; Ex_07.2 notebooks 1 - 3*

2 What do Poisson's and Helmholtz's equations describe, and what must a Helmholtz network be protected from?

Poisson is a steady balance between a source and diffusion. Helmholtz adds a term in the field itself, so it has modes, and $u \equiv 0$ always satisfies it: the network needs a unit-norm term and a floor on λ . In the slot, λ reaching the first eigenvalue is thermal runaway. *Liu ch. 2.4, 6.2; Ex_07.2 notebook 2*

3 - 4 • IN TIME

3 What does time add: how do the heat and the wave equation treat a disturbance, and an error made early?

Heat spreads and forgets: every feature decays, the sharpest fastest, and an early error is damped. A wave travels at a finite speed and keeps its shape, so an early error is carried along. Heat needs one initial condition, the wave two. *Liu ch. 4.1, 5.1; Ex_07.2 notebooks 1, 3*

4 How does a PINN meet the conditions of each type, and which type is hardest for it?

A condition with a formula is built into the trial function: the walls, the initial field. One without, like the panel's initial velocity, is a term in the loss. The wave is hardest: its error is not damped, and its residual must be scaled before the network moves. *Liu ch. 4.2, 5.2; Ex_07.2 notebook 3*

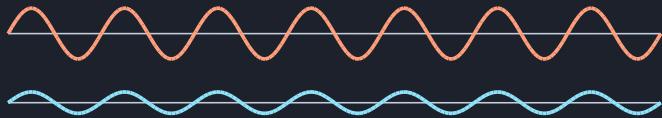
Heat

Heat — energy that flows because of a temperature difference. In a solid each atom vibrates about a fixed place; the temperature at a point is the average energy of that vibration. A hot atom shakes its neighbours through the bonds: the energy moves on, the atoms stay.

AT THE SCALE OF ATOMS

- **temperature** — at a point, an average over many atoms
- **frequency** — about 10^{13} Hz; heating widens the swing
- **bonds** — electric forces pass the push on, no radiation
- **metals** — free electrons carry heat as well

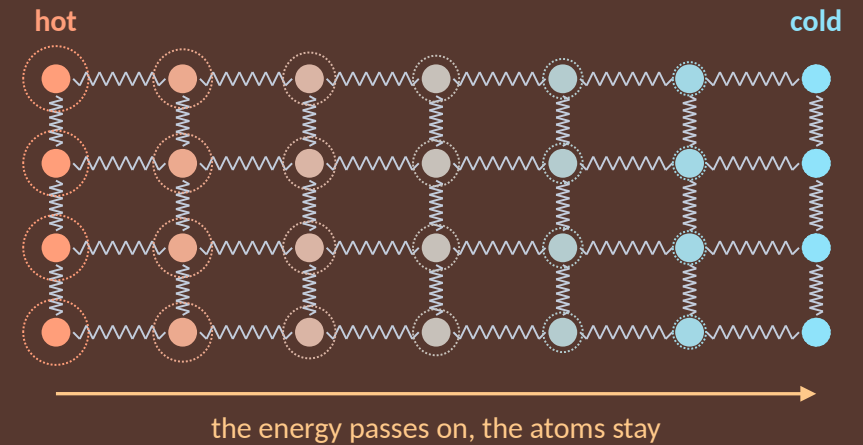
ONE ATOM AGAINST TIME



hot: a wide swing

cold: a narrow swing

ATOMS IN A SOLID, VIBRATING



Each dot is an atom and each spring a bond. The dotted ring is how far the atom swings: wide where it is hot, narrow where it is cold.

Transport Analogies

Transport analogy — heat, charge, magnetic flux, a fluid and a diffusing species all flow the same way: a difference drives a flux, a property of the material sets it, and the path opposes it with a resistance. The difference equals the resistance times the flow, an Ohm's law.

NOT A FLOW: A STRETCHED BAR

$\delta = F \cdot L / (E \cdot A)$ has the same form, but the bar stores energy, like a capacitor.

SIX FLOWS, ONE LAW

	heat	current	magnetic flux	fluid in a pipe	diffusion	charge transfer
driven by	ΔT [K]	ΔV [V]	NI [A]	Δp [Pa]	Δc [mol/m ³]	η [V]
flux	q [W/m ²]	J [A/m ²]	B [T]	u [m/s]	j [mol/(m ² s)]	i_n [A/m ²]
law	$q = -k \nabla T$	$J = -\sigma \nabla V$	$B = \mu H$	$u = -(r^2/8\rho\nu) \nabla p$	$j = -D \nabla c$	$i_n \approx i_0 \eta / V_T$
resistance	$R_{th} = L / (k \cdot A)$	$R = L / (\sigma \cdot A)$	$\mathcal{R} = L / (\mu \cdot A)$	$R_h = 8\rho\nu L / (\pi r^4)$	$R_D = L / (D \cdot A)$	$R_{ct} = V_T / (i_0 \cdot A)$
Ohm's law	$\Delta T = R_{th} P$	$\Delta V = R I$	$NI = \mathcal{R} \Phi$	$\Delta p = R_h Q$	$\Delta c = R_D I / F$	$\eta = R_{ct} I$
unit of R	[K/W]	[Ω]	[A/Wb]	[Pa·s/m ³]	[s/m ³]	[Ω]
mass carried	none	none, net	none	yes, the fluid	yes, the species	yes, the ions
discussed in	L8.1 Stationary Heat L8.2 Transient Heat	L10.1 Ionic Diffusion & Charge Conservation	not in this course	L9.1 Laminar Flow L9.2 Turbulent Flow	L10.1 Ionic Diffusion & Charge Conservation	L10.1 Ionic Diffusion & Charge Conservation

k, σ, μ, D thermal and electrical conductivity, permeability, diffusivity, ρ, ν density, kinematic viscosity, c concentration, η overpotential, i_0 exchange current density
 L, A, r length, cross-section, radius of the path, u mean speed, P, I, Φ, Q the total flows, F Faraday's constant, $V_T = RT/F$ the thermal voltage, 25.7 mV at 25 °C

Conduction

Conduction — heat carried through a material by its atoms and electrons, with no motion of the material itself. It flows from hot to cold, at a rate proportional to the temperature gradient: Fourier's law, with the conductivity k as the property of the material.

FOURIER'S LAW

$$\mathbf{q} = -k \nabla T$$

$\mathbf{q}$ the heat flux [W/m^2], k the conductivity [$\text{W}/(\text{m}\cdot\text{K})$], ∇T the gradient [K/m]

- **driven by** — a temperature gradient, inside a solid or a still fluid
- **direction** — from hot to cold, hence the minus sign
- **examples** — a chip into its heat sink, a winding into its slot

CONDUCTION, DRAWN



the solid, hot to cold

q runs down the gradient

CONDUCTIVITY OF COMMON MATERIALS

material	k [$\text{W}/(\text{m}\cdot\text{K})$]
copper	400
aluminium	237
silicon	150
stainless steel	15
glass	1
water	0.61
air	0.03

Typical values at room temperature. Four orders of magnitude lie between copper and air: a heat sink is metal, and insulation is trapped air.

Convection

Convection — heat carried away from a surface by a moving fluid. The motion of the fluid is not solved for: its effect is put into one coefficient h , and the flux is proportional to the difference between the surface and the fluid. That is Newton's law of cooling.

NEWTON'S LAW OF COOLING

$$q = h(T_s - T_\infty)$$

h the heat transfer coefficient [W/(m²K)], T_s the surface, T_∞ the fluid far from it

FROM THE SPEED TO THE COEFFICIENT

$$Re = uL/\nu, \quad Nu = CRe^m Pr^{1/3}, \quad h = Nu k_f / L$$

Re , Nu , Pr the Reynolds, Nusselt and Prandtl numbers, without units, $Pr = \nu/\alpha$; C , m set by the flow
 h [W/(m²K)], L the length of the wall along the flow [m], u the fluid's speed [m/s]

k_f the fluid's conductivity [W/(m·K)], ν , α its kinematic viscosity and thermal diffusivity [m²/s]

CONVECTION, DRAWN



the fluid, at T_∞ , moves past
 the surface, at T_s , gives up q

HEAT TRANSFER COEFFICIENT

flow	h [W/(m ² K)]
natural, air	2 – 25
forced, air	25 – 250
natural, water	50 – 1 000
forced, water	100 – 20 000
boiling water	2 500 – 100 000

Air at 2 m/s along a 0.1 m fin: $Re = 12\,500$, $Nu = 66$, $h = 17$ W/(m²K). At 4 m/s, $h = 24$ W/(m²K). Laminar flow ($C = 0.664$, $m = 1/2$) gives $h \propto u^{0.5}$; turbulent flow ($C = 0.037$, $m = 4/5$) gives $h \propto u^{0.8}$.

Radiation

Radiation — heat exchanged as electromagnetic waves between surfaces that see each other; it needs no material in between. The power a surface radiates grows with the fourth power of its absolute temperature: the Stefan–Boltzmann law, with the emissivity ε of the surface.

STEFAN-BOLTZMANN LAW

$$q = \varepsilon \sigma (T_s^4 - T_{\text{sur}}^4)$$

ε the emissivity, 0 to 1, σ the Stefan-Boltzmann constant, $5.67 \times 10^{-8} \text{ [W/(m}^2\text{K}^4\text{)]}$
 q the heat flux $[\text{W/m}^2]$, T_s , T_{sur} the surface and the surroundings $[\text{K}]$

- **needs** — no medium: it crosses a vacuum
- **nonlinear** — the fourth power, with T in kelvin
- **examples** — the sun, a furnace wall, a radiator in space

RADIATION, DRAWN



the hot surface radiates
the surroundings radiate back

EMISSIVITY OF COMMON SURFACES

surface	ε
polished aluminium	0.05
anodised aluminium	0.80
black paint	0.95
a black body	1

A black body, $\varepsilon = 1$, absorbs all the radiation that falls on it and radiates the most any surface can, σT^4 ; a real surface radiates the fraction ε of that. At 80°C with $\varepsilon = 0.90$, in surroundings at 25°C , it radiates 390.50 W/m^2 , of one size with natural convection (550 W/m^2 at $h = 10 \text{ W/(m}^2\text{K)}$).

Heat Into a Solid

Heat into a solid — heat from outside reaches the surface by contact, through a fluid or as radiation; inside the solid it is transferred by conduction alone. So conduction is the field equation a PINN solves, and convection and radiation enter as boundary conditions.

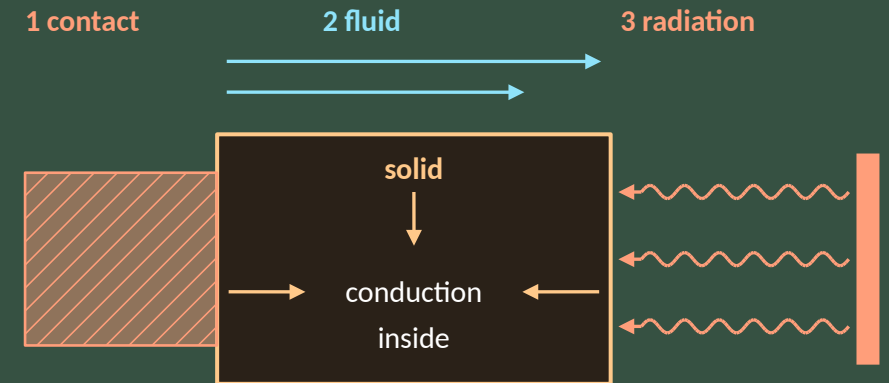
WHERE EACH MECHANISM ENTERS

	acts	in the PINN
conduction	inside the solid	the equation the PINN solves
convection	at a surface, into a fluid	a Robin wall, with h given
radiation	between surfaces	left out, or counted in h

WHY THE OTHER TWO ARE NOT SOLVED

- **convection** — its field is the flow of the fluid: that is L9
- **radiation** — T^4 and the view between surfaces: nonlinear
- **both** — act at the wall, so a boundary condition carries them

HEAT FROM OUTSIDE, THREE ROUTES



Every route ends at the surface: radiation is absorbed in a thin layer there. Inside, it is transferred by conduction. Radiation is not small: at 80 °C against 25 °C it acts like $h = 7.10 \text{ W}/(\text{m}^2\text{K})$.

Steady Conduction

Steady conduction — Fourier's law in an energy balance: in steady state what leaves a small volume equals what is generated in it, $\nabla \cdot \mathbf{q} = Q$. With $\mathbf{q} = -k\nabla T$ that is Poisson's equation for the temperature, with Q/k as its source.

FOURIER'S LAW IN AN ENERGY BALANCE

$$\mathbf{q} = -k \nabla T, \quad \nabla \cdot \mathbf{q} = Q$$

$$T_{xx} + T_{yy} + \frac{Q}{k} = 0$$

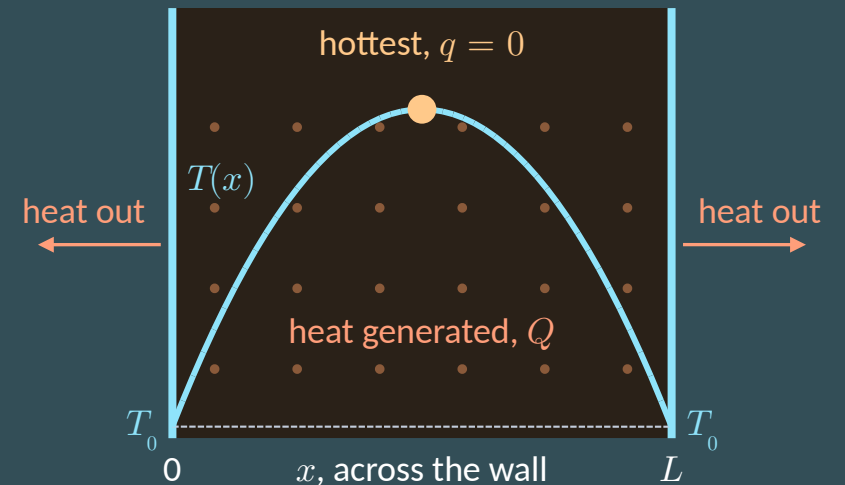
$\mathbf{q}$ the heat flux, k the conductivity [W/(m·K)]

Q the heat generated [W/m³]

CONDUCTION IN PYTORCH

```
T = net(xy)           # the temperature, in kelvin
q = -k * grad(T, xy)   # Fourier's law; grad = autograd
F = div(q, xy) - Q      # the energy balance, zero inside
```

A WALL THAT GENERATES HEAT



A plate heated inside, by a current say, with both faces cooled to T_0 . The temperature is a parabola, hottest in the middle, where the flux is zero: half the heat leaves through each face.

Variable Conductivity

Variable conductivity — when k changes with position it stays inside the divergence: $\nabla \cdot (k \nabla T) + Q = 0$. Across an interface between two materials the gradient jumps and the normal flux does not, so the divergence must not be expanded; autograd differentiates $k \nabla T$ as written.

K INSIDE THE DIVERGENCE

$$\nabla \cdot (k(\mathbf{x}) \nabla T) + Q(\mathbf{x}) = 0$$

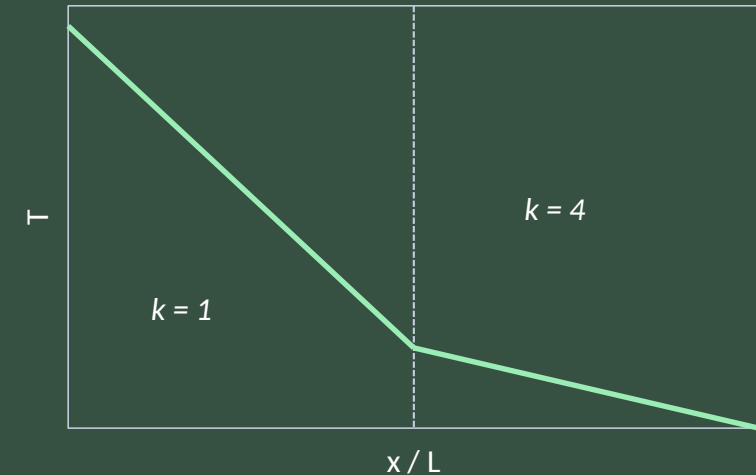
$k(\mathbf{x})$ the conductivity at each point, $k \nabla T$ differentiated as one quantity

- **at an interface** — the slope jumps, the flux does not
- **in code** — differentiate the product, not its parts

VARIABLE k IN PYTORCH

```
k = k_field(xy)           # the conductivity at each point
q = -k * grad(T, xy)      # Fourier's flux, one quantity
F = div(q, xy) - Q        # differentiated as written
```

TWO MATERIALS, ONE INTERFACE



Left face at 1, right face at 0, no generation. The slope falls fourfold at the interface; the flux $k dT/dx$ is the same on both sides.

Thermal Boundary Conditions

Thermal boundary conditions — three ways to fix a wall: a temperature (Dirichlet), a heat flux (Neumann), or convection to a fluid (Robin), which ties the flux to the wall temperature. Only a temperature can be built into the network; the other two stay loss terms.

A TEMPERATURE, A FLUX, A CONVECTION

Dirichlet

$$T = \bar{T} \text{ on } \partial\Omega_D$$

Neumann

$$-k \frac{\partial T}{\partial n} = \bar{q} \text{ on } \partial\Omega_N$$

Robin

$$-k \frac{\partial T}{\partial n} = h(T - T_\infty)$$

$\bar{T}$ the wall temperature, $\bar{q}$ the wall heat flux, n the outward normal
 h the heat transfer coefficient, T_∞ the fluid temperature

WHAT EACH CONDITION FIXES AT THE WALL



ROBIN • CONVECTION AT THE WALL

- **balance** — heat conducted to the wall equals heat the fluid carries away
- h — heat transfer coefficient [W/(m²K)], or film coefficient: the fluid layer at the wall in one number
- **radiation** — where it matters, it is counted in h

BIOT NUMBER

$$\text{Bi} = \frac{hL}{k} = \frac{L/k}{1/h}$$

L the size of the body, L/k resists conduction inside
 $1/h$ resists convection at the wall

$\text{Bi} \ll 1$: nearly insulated (Neumann). $\text{Bi} \gg 1$: the wall is at T_∞ (Dirichlet). 1 mm of silicon in still air: 1.67×10^{-4} .

Well-Posedness

Compatibility condition — a problem with a flux condition on every wall has a solution only if the heat out through the walls equals the heat generated, and then only up to a constant: any shift of T fits the same equation and walls. A PINN picks one without warning.

HEAT OUT EQUALS HEAT GENERATED

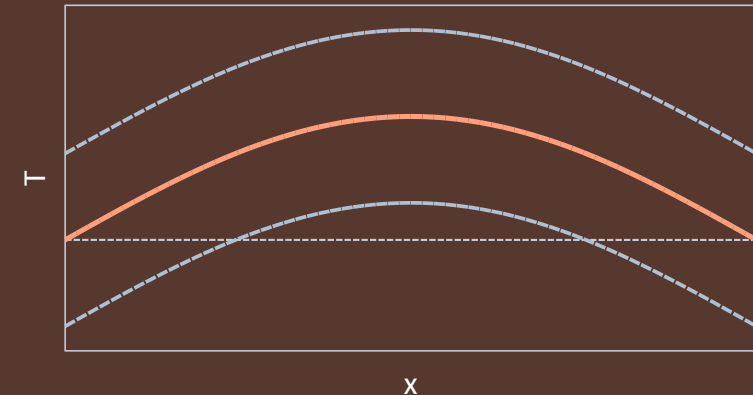
$$\oint_{\partial\Omega} \bar{q} \, ds = \int_{\Omega} Q \, d\Omega$$

$\bar{q}$ the flux out through the boundary $\partial\Omega$, Q the heat generated inside

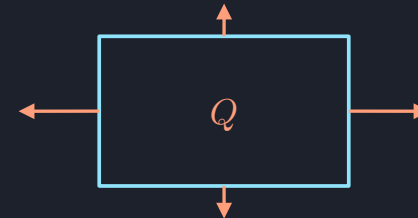
- **one value pinned** — fixes the constant
- **one wall at a temperature** — fixes it too

TALK • 2 MIN How would you explain why a problem with only flux conditions has no single solution, in your own words? Then, in the plot on the right, which of the three curves is right, and what would you add to the loss to choose it?

THE SAME FIELD, SHIFTED



THE BALANCE, DRAWN



heat generated inside
= heat out through the walls
Only then is there a steady field,
and it can still be shifted up or down.

Non-Dimensional Form

Non-dimensional form — the problem rewritten in scaled variables: lengths divided by the device size L , temperature by a rise $\Delta T = QL^2/k$. Every term is then of order one, as the initialisation and the loss weights assume. Train scaled, and convert back at the end.

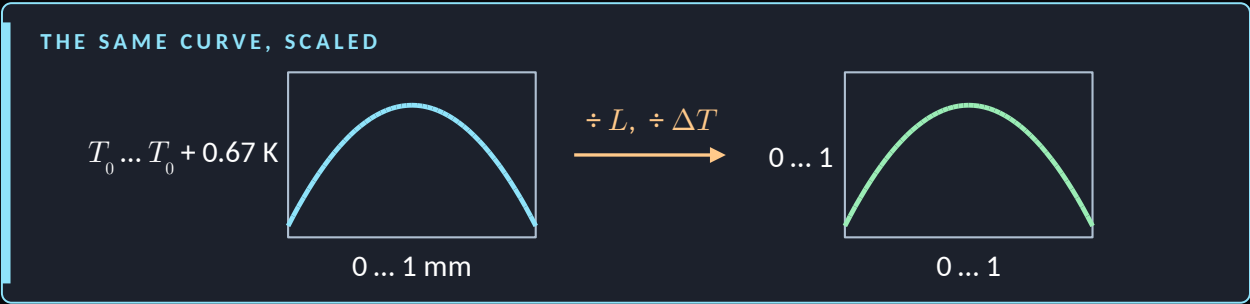
SCALE, TRAIN, CONVERT BACK

$$\theta = \frac{T - T_0}{\Delta T}, \quad \xi = \frac{x}{L}, \quad \Delta T = \frac{QL^2}{k}$$

θ the scaled temperature, ξ the scaled coordinate, ΔT the temperature scale

$$\theta_{\xi\xi} + \theta_{\eta\eta} + 1 = 0$$

$\eta = y/L$; the same equation, every coefficient equal to one



A SILICON DEVICE, BEFORE AND AFTER

	physical	scaled
length L [mm]	1	1
conductivity k [W/(m·K)]	150	1
source Q [W/m³]	10^8	1
rise $\Delta T = QL^2/k$ [K]	0.67	1
coefficient h [W/(m²K)]	25	$Bi = 1.67 \times 10^{-4}$

In metres and kelvin the source term is ten to the eighth and the length ten to the minus third. Scaled, every term is of order one.

Sampling a Real Domain

Rejection sampling — points drawn uniformly over a box around the domain, keeping those the level set places inside, $\varphi > 0$. On a curved wall the points are spaced by arc length rather than by angle, so the ends of the hole, where it bends most, get their share.

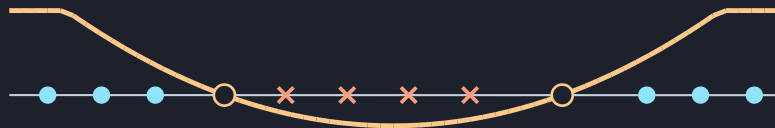
ONE FORMULA: INSIDE, OUTSIDE, ON THE WALL

$$\phi(x, y) = \left(\frac{x-x_c}{a}\right)^2 + \left(\frac{y-y_c}{b}\right)^2 - 1$$

(x_c, y_c) the centre, a, b the half axes; $\phi < 0$ in the hole, $\phi = 0$ on its wall

- **inside** — keep a point where $\varphi > 0$
- **on the wall** — equal steps of arc length

THE LEVEL SET ALONG A LINE THROUGH THE HOLE

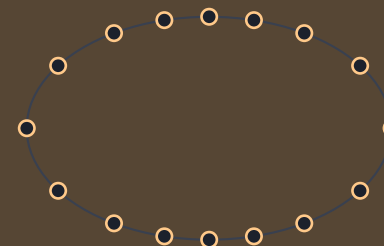
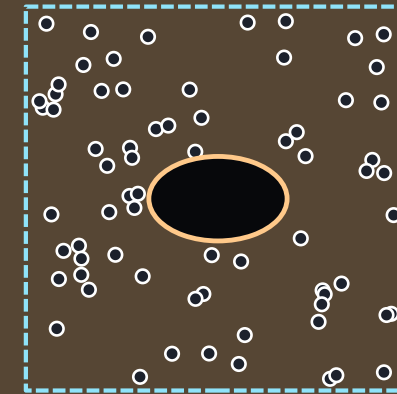


$\varphi > 0$: in the plate, kept

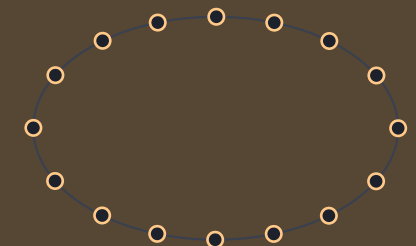
$\varphi = 0$: the hole's wall

$\varphi < 0$: in the hole, rejected

THE PLATE, AND ITS WALL POINTS



equal angle: ends bare



equal arc length

Boundary Multiplier

Boundary multiplier — a function that is zero on a boundary and positive in the domain. Multiplied into the network's output, it makes a value hold there exactly. For an ellipse the level set ϕ is one; for a shape with no formula, a small second network is trained to be one, then frozen.

THE HOLE WALL, BUILT IN

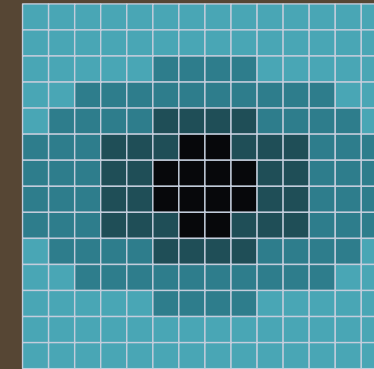
$$T(\mathbf{x}) = \phi(\mathbf{x}) \mathcal{N}_{\hat{w}}(\mathbf{x})$$

ϕ the level set, zero on the hole; $\mathcal{N}_{\hat{w}}$ the network; $T = 0$ there for any weights

- **a value** — held exactly by a factor
- **a derivative** — cannot be, stays a loss term

TALK • 2 MIN How would you explain what a boundary multiplier does, in your own words? Then, in the figure on the right, where is the multiplier small, and what does that do to the network's gradient near the hole?

ϕ OVER THE PLATE



dark: the hole

FROM THE WALL INTO THE PLATE



Flux on a Curved Boundary

Normal derivative — the rate of change of T across a wall, $\partial T / \partial n = \nabla T \cdot \mathbf{n}$, where $\mathbf{n}$ is the unit normal pointing out of the material. On a curved wall $\mathbf{n}$ changes with position: it comes from the gradient of the level set, and its sign decides whether heat leaves or enters.

THE NORMAL FROM THE LEVEL SET

$$\frac{\partial T}{\partial n} = \nabla T \cdot \mathbf{n}, \quad \mathbf{n} = -\frac{\nabla \phi}{|\nabla \phi|}$$

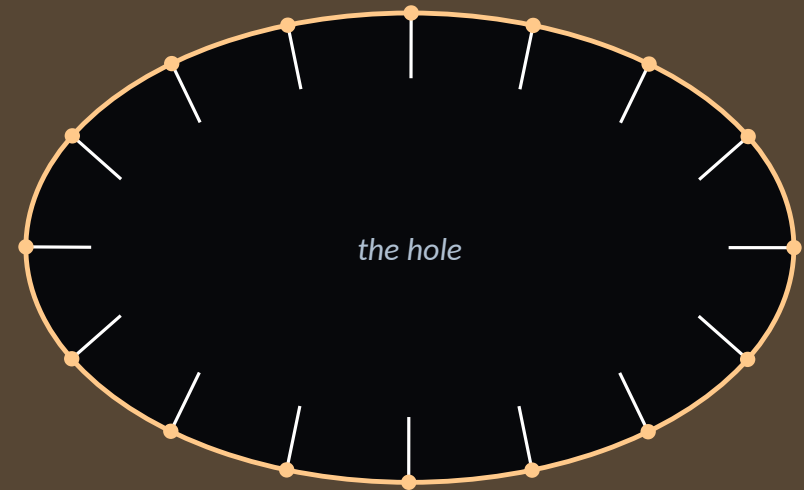
$\mathbf{n}$ the unit normal out of the plate; minus, because out of the plate is into the hole

- **on an edge** — the normal is a constant
- **on the hole** — it turns with the wall

THE NORMAL IN PYTORCH

```
g = grad(phi(xy_h), xy_h)           # the level set's gradient
n = -g / g.norm(dim=1, keepdim=True) # unit normal, out of the plate
q_n = -k * (grad(T, xy_h) * n).sum(1) # Fourier's flux through the wall
```

NORMALS ON THE HOLE WALL



Each white stick is $\mathbf{n}$ at an amber point: it turns with the wall and always points out of the plate. A sign error inverts the flux, not the magnitude.

Manufactured Solution

Manufactured solution — a solution chosen in advance, with the source derived from it by differentiation, so the answer is known exactly. Solving that problem checks the residual, the trial solution and the training loop before they meet a problem with no answer to compare against.

CHOOSE THE ANSWER, DERIVE THE SOURCE

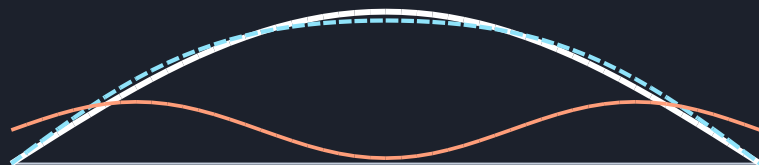
$$T = \sin(\pi x)\sin(\pi y) \Rightarrow \frac{Q}{k} = 2\pi^2 \sin(\pi x)\sin(\pi y)$$

T the chosen answer, zero on the four edges; Q/k the source derived from it

$$\varepsilon = \frac{\|T_{\hat{w}} - T_{\text{exact}}\|_2}{\|T_{\text{exact}}\|_2}$$

$T_{\hat{w}}$ the network's answer, T_{exact} the chosen one

THE ANSWER IS KNOWN, SO THE ERROR IS A CURVE

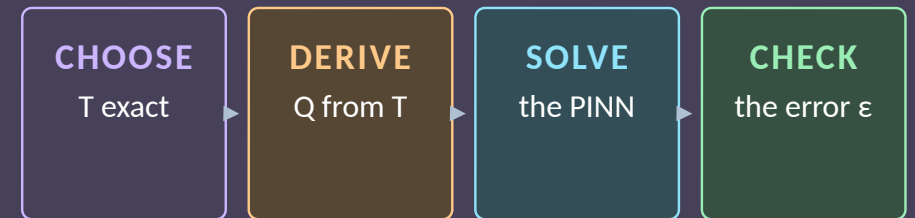


the chosen answer

the network's

their difference, enlarged

FOUR STEPS BEFORE THE PLATE



what it checks

the residual, the trial, the loop

what it cannot

the geometry of the plate

An error found here costs minutes; the same error found on the plate costs an afternoon.

Heated Plate with a Hole

Heated plate with a hole — the case study of this block: a square plate with uniform heat generation, cooled through an elliptical hole held at $T = 0$, with insulated outer edges, $\partial T / \partial n = 0$. The hole wall is built into the network by its level set; the edges stay a loss term.

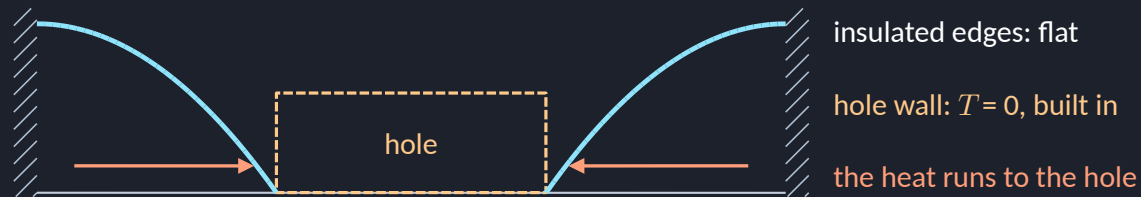
ONE BUILT-IN WALL, ONE LOSS TERM

$$\mathcal{L} = \mathcal{L}_{PDE} + w_N \mathcal{L}_{NBC}$$

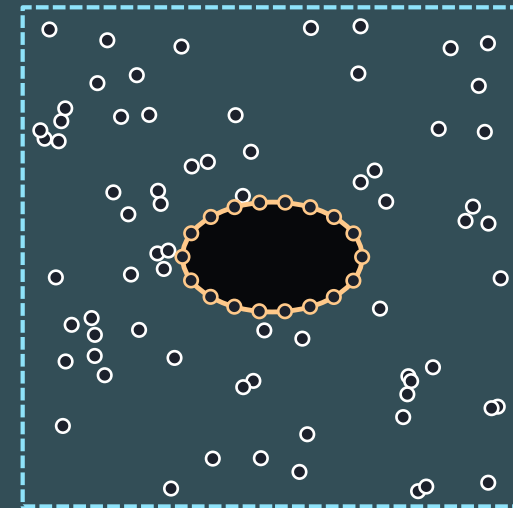
$\mathcal{L}_{PDE}$ the residual inside, $\mathcal{L}_{NBC}$ the insulated edges, w_N their weight

- **the hole** — $T = 0$, by the multiplier
- **the edges** — no flux, a weighted term

THE TEMPERATURE ALONG A LINE THROUGH THE HOLE



THE PLATE OF EX_08.1



amber wall

$T = 0$, built in

dashed edges

$\partial T / \partial n = 0$, a loss term

Flux Weight

Flux weight — the weight w_N on the insulated-edge term. Too small, and the network ignores the edges and heat leaks through them; too large, and the residual inside pays for a perfect wall. It is chosen by a sweep that watches both terms, not the total loss.

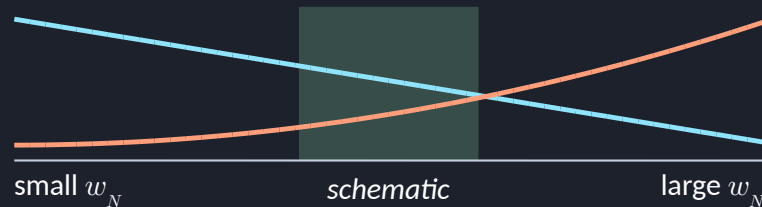
ONE RUN PER WEIGHT, BOTH TERMS WATCHED

$$\mathcal{L} = \mathcal{L}_{PDE} + w_N \mathcal{L}_{NBC}$$

$\mathcal{L}_{PDE}$ the residual inside, $\mathcal{L}_{NBC}$ the insulated edges, w_N their weight

- **the total loss** — hides which term is paying
- **the choice** — a trade, argued from both

ONE TERM FALLS, THE OTHER RISES



the edge term: falls
the residual inside: rises
between: argued from both

WHAT EACH END OF THE SWEEP DOES

w_N small

the edges leak heat

w_N large

the residual inside pays

between

argued from both terms

The weight is a rate of exchange between a flux and a residual, which have different units. Scaling changes the rate before any sweep does.



Energy Balance Check

Energy balance check — a check on the trained network, not a new law. Fourier's law gives the heat flux at each point of the hole's wall; added up around the hole, that is the heat leaving the plate. In steady state it equals the heat generated: the energy balance, for the whole plate.

FOURIER'S FLUX, ADDED UP AROUND THE HOLE

$$P_h \overline{q_n} = Q A_\Omega$$

P_h the hole's perimeter, $\overline{q_n}$ the mean of Fourier's flux into it, A_Ω the plate's area

- **at a point** — Fourier's law gives the flux into the hole
- **around the hole** — its mean times the perimeter is the heat out
- **in the plate** — Q times the area is the heat generated

THE BALANCE IN PYTORCH

```
q_n = -k * (grad(T, xy_h) * n).sum(1)    # 1: Fourier's flux at each wall point
out = q_n.mean() * perimeter              # 2: heat out; points evenly spaced
gen = Q * (1 - pi * a * b)                # 3: heat generated; compare with out
```

THE CHECK IN EX_08.1

	value
heat generated [W/m]	9378
heat out [W/m]	9332
heat out / generated [%]	99.51
out short by [%]	0.49

The plate with a hole has no exact solution to compare against, so this balance is a check with a physical reason to come out right: heat generated is known, heat out is the network's. Per metre of depth. Whether 0.49 % is small enough is your decision.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
energy method	an energy, first derivatives only	Liu ch. 2.8	elasticity, structures
triangulated domain	points and weights from a mesh	Liu ch. 3.2	any drawn geometry
convergence study	error against points and size	Liu ch. 4.9.3	a design sign-off
<u>diffusion surrogate</u>	trained on solved cases, then steered	UDL ch. 14 – 18	many boundary conditions
PINN or FEM	when each is the better tool	Liu ch. 2.10	one-off steady solves

WHY THEY ARE LEFT OUT

The plate of Ex_08.1 needs none of them: its hole has a formula, its answer can be checked by an energy balance, and it is solved once. A finite-element solver would solve it too; the method earns its place on the inverse problems of L8.2.

Exercise 8.1

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the plate, its data, its physics, scaled
- **4** — a fitted reference, and finite differences
- **5** — a PINN with the hole built in; the three compared
- **6** — what the comparison says
- **7 – 8** — the report; mini projects MP8.1A and MP8.1B

THE QUESTION EACH PART ANSWERS

- 1 – 3** how is the plate posed?
- 4** what is the answer, to 0.10 K?
- 5** how does a PINN compare?

REPORT QUESTION, TO CONCLUDE

For this one plate, solved once, which of the three methods would you choose, and what would have to change for the network to be worth its training?

Key Takeaways

HEAT TRANSFER

what it is	conduction, convection, radiation
function	T by conduction; the rest are walls
train, tune	scale first; flux walls weighted
vs Poisson	units, k inside, three walls
used for	devices, plates, heat sinks

REAL GEOMETRY

what it is	a domain with no polynomial factor
function	ϕ : inside, multiplier, normal
train, tune	arc length on the wall; w_N
vs rectangle	n and the factor vary with x
used for	plates with holes, device sections

Next — L8.2: the same plate, in time.

Questions

1 – 2 • HEAT TRANSFER

- 1** What are the three mechanisms of heat transfer, and which of them does a PINN solve in L8?

Conduction through a material (Fourier's law), convection from a surface into a moving fluid (Newton's law of cooling, with h), and radiation between surfaces (Stefan–Boltzmann, T^4). Only conduction is a field equation inside the solid; the other two act at the surface and enter as boundary conditions. *Incropera ch. 1*

- 2** What does a steady conduction problem need before a PINN solves it, and how is the result checked?

Fourier's law in an energy balance gives Poisson's equation, with k inside the divergence. Each wall carries a temperature, a flux or convection; with a flux on every wall the level is not fixed. Scale it so every term is of order one, then check by a manufactured solution and the energy balance. *Liu ch. 2.2, 2.4, 4.4; Ex_08.1 sections 3 – 5*

3 – 4 • REAL GEOMETRY

- 3** How is a domain with a hole sampled, and how is a temperature built into its wall?

Points are drawn over a box and kept where the level set ϕ is positive; on the curved wall they are spaced by arc length. A boundary multiplier, zero on the wall and positive inside, multiplies the network's output, so the wall temperature holds exactly. *Liu ch. 3.1 – 3.3; Ex_08.1 section 5*

- 4** How is a flux condition imposed on a curved wall, and how is its weight chosen?

As a loss term on the normal derivative, the gradient of T along the outward unit normal, which comes from the level set. Its weight is found by a sweep that watches the flux term and the residual separately: too small and heat leaks through the wall, too large and the interior pays. *Liu ch. 2.2.3, 3; Ex_08.1 section 5*

NEXT

L8.2 — Transient Heat

The same plate, in time: an initial field, causality, and a diffusivity recovered from sensors.

BEFORE THE NEXT SESSION

read	Liu ch. 3, 4.6
run	Ex_08.1, sections 1 to 6
bring	the energy balance, and your error
think about	which of your walls can be built in

The steady plate is solved and checked. Next it heats up.